

2

NUMBER AND ALGEBRA

GRAPHING LINES

Straight lines are an important part of our environment. We play sport on courts using parallel and perpendicular lines, and skyscrapers would not be standing without straight lines. We can also use straight lines to model different types of data and predict future outcomes.



Chapter outline

	Working mathematically				
2.01 Length, midpoint and gradient of an interval	U	F	PS	R	C
2.02 Parallel and perpendicular lines	U	F			
2.03 The gradient–intercept equation $y = mx + c$	U	F		R	C
2.04 The general equation $ax + by + c = 0$*	U	F		R	C
2.05 The point–gradient equation $y - y_1 = m(x - x_1)$*	U	F		R	C
2.06 Finding the equation of a line	U	F		R	C
2.07 Applying linear equations	U	F		R	C
2.08 Equations of parallel and perpendicular lines	U	F		R	C
2.09 Coordinate geometry proofs*	U	F	PS	R	C

***PATH**

Wordbank

general form Any linear equation expressed as $ax + by + c = 0$, where a, b and c are integers and a is positive

gradient The steepness of a line or interval, measured by the fraction $\frac{\text{rise}}{\text{run}}$

gradient–intercept form Any linear equation expressed as $y = mx + c$, where m is the gradient and c is the y-intercept

linear equation An equation whose graph is a straight line

parallel lines Lines that point in the same direction and have the same gradient

perpendicular lines Lines that cross at right angles (90°) and have gradients whose product is −1

point-gradient form Any linear equation expressed as $y - y_1 = m(x - x_1)$, where m is the gradient and (x_1, y_1) is a point on the line

***x*-intercept** The x-value at which a graph cuts the x-axis

***y*-intercept** The y-value at which a graph cuts the y-axis

Quiz
Wordbank 2

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

Videos (16):

2.01 The gradient of a line • Coordinate geometry • Distance, midpoint and gradient formulas
2.02 Gradients of perpendicular lines
2.03 Graphing linear equations • Testing if a point lies on a line • Horizontal and vertical lines • Graphing $y = mx + c$ • Gradient and y-intercept of a line • The gradient-intercept formula
2.04 The general equation $ax + by + c = 0$
2.05 The point–gradient formula
2.06 Finding the equation of a line
2.07 Applying linear functions
2.08 Equations of parallel lines • Equations of perpendicular lines

PhET interactives (4):

2.03 Graphing lines • Function builder: Basics • Function builder • Graphing slope-intercept

Quizzes (5):

- Wordbank 2
- SkillCheck 2
- Mental skills 2
- Language of maths 2
- Test yourself 2

Skillsheets (2):

SkillCheck Pythagoras' theorem
2.03 Graphing linear equations

Worksheets (12):

2.01 Gradient, midpoint, distance • Gradient between 2 points
2.03 A page of lines • A page of number planes • Graphing linear equations • Gradient linear equations (Advanced) • Straight-line equations
2.04 Parallel and perpendicular lines
2.06 Finding the equation of a line • A page of lines
2.08 Writing equations of lines
2.09 Geometry problems using coordinates
Mind map Graphing lines (Advanced)

Puzzles (10):

2.01 Intervals match-up • Finding coordinates for given segment lengths
2.02 Gradients of parallel and perpendicular lines
2.03 Equation of a line • Equations in gradient form
2.04 Linear equations code puzzle
2.06 Straight-line equations
2.08 Linear equations match-up • Equations of parallel lines
Language of maths Graphing lines crossword

Technology (1):

2.01 Midpoint and distance between 2 points

Spreadsheets (2):

2.01 Midpoint and distance between 2 points
2.03 Drawing straight lines: $y = mx + c$

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- find the length, midpoint and gradient of an interval on a number plane
- (PATH) use *formulas* to find the length, midpoint and gradient of an interval
- (PATH) find the angle of inclination θ of a line using the formula $m = \tan\theta$
- find the properties of the gradients of parallel and perpendicular lines
- (PATH) find the x- and y-intercepts of a line from its equation, then graph the line
- test whether a point lies on a line
- use the gradient–intercept equation of a straight line $y = mx + c$
- (PATH) convert between the general form of a linear equation $ax + by + c = 0$ and the gradient–intercept equation

- (PATH) use the point–gradient form of a linear equation $y - y_1 = m(x - x_1)$ to find the equation of a line given the gradient and a point, or 2 points
- find the equation of a line from its graph
- solve problems that can be modelled by linear equations
- find the equation of a line that is parallel or perpendicular to a given line
- (PATH) prove geometrical properties using coordinate geometry methods

SkillCheck ANSWERS ON P. 609

1 For this number plane, find:

a the midpoint of interval BC

b the midpoint of interval HE

c the length of interval GC

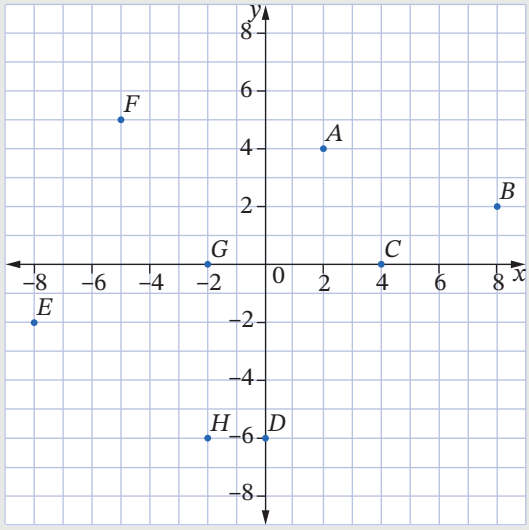
d the length of interval GH

e the lengths of AC and BC, correct to one decimal place

f the type of triangle $\triangle ABC$ is

g the gradient of GE

h the gradient of EH



Quiz
SkillCheck 2

Skillsheet
Pythagoras' theorem

Length, midpoint and gradient of an interval

2.01

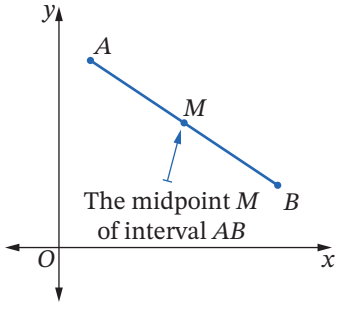
The **length** of an interval AB (or the **distance** between A and B) can be calculated using Pythagoras' theorem.

The **midpoint** of AB is halfway between A and B.

- Its x-coordinate is the average of the x-coordinates of A and B.
- Its y-coordinate is the average of the y-coordinates of A and B.

The **gradient** of an interval measures its steepness.

$$m = \frac{\text{vertical rise}}{\text{horizontal run}} = \frac{\text{rise}}{\text{run}}$$



Video
The gradient of a line

Worksheets
Gradient, midpoint and distance

Gradient between 2 points

Puzzle
Intervals match-up

Technology
Midpoint and distance between 2 points

Spreadsheet
Midpoint and distance between 2 points

Example 1

For the interval joining $P(-5, 8)$ and $Q(3, 6)$, find:

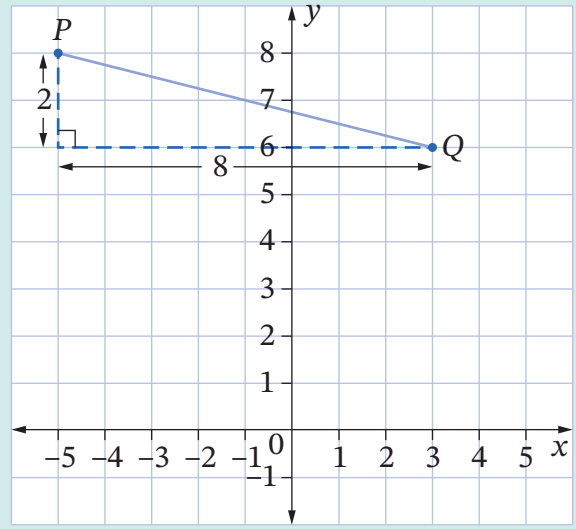
a its length, correct to one decimal place

b its midpoint

c its gradient.

Solution

a Draw a right-angled triangle on the number plane with PQ as the hypotenuse.



$PQ^2 = 2^2 + 8^2$

$= 68$

$PQ = \sqrt{68}$

$= 8.2462...$

≈ 8.2 units

b For $P(-5, 8)$ and $Q(3, 6)$, the average of the x-coordinates is $\frac{-5+3}{2} = -1$.

The average of the y-coordinates is $\frac{8+6}{2} = 7$.

From the diagram in part **a**, a midpoint at $(-1, 7)$ looks reasonable.

$\therefore$ The midpoint of PQ is $(-1, 7)$.

c $m = \frac{\text{rise}}{\text{run}}$

$= \frac{-2}{8}$

$= -\frac{1}{4}$

The gradient is negative, the line slopes downward.

The distance, midpoint and gradient formulas

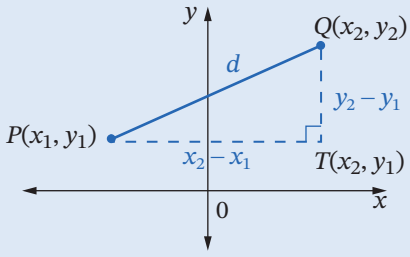
For an interval joining $P(x_1, y_1)$ and $Q(x_2, y_2)$:

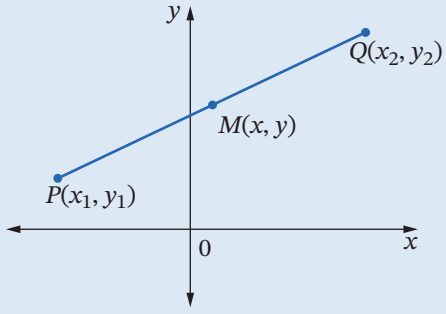
Distance $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Midpoint $M(x, y) \equiv \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

We've used '$\equiv$' ('is identical to') rather than '$=$' because we are referring to the point (x, y), not a number.

Gradient $m = \frac{y_2 - y_1}{x_2 - x_1}$





Example 2

For the interval joining $P(-5, 8)$ and $Q(3, 6)$ from Example **1**, use a formula to find:

a its length, correct to one decimal place

b its midpoint

c its gradient.

Solution

For $P(-5, 8)$ and $Q(3, 6)$: $x_1 = -5, y_1 = 8, x_2 = 3, y_2 = 6$.

(x_1, y_1) (x_2, y_2)

a $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$= \sqrt{(3 - (-5))^2 + (6 - 8)^2}$

$= \sqrt{68}$

$= 8.2462\ldots$

≈ 8.2 units

b $M(x, y) \equiv \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

$\equiv \left(\frac{-5 + 3}{2}, \frac{8 + 6}{2}\right)$

$\equiv (-1, 7)$

c $m = \frac{\text{difference in } y}{\text{difference in } x}$

$= \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{6 - 8}{3 - (-5)}$

$= \frac{-2}{8}$

$= -\frac{1}{4}$

PATH

Video Coordinate geometry

2.01

Videos Distance, midpoint and gradient formulas

Coordinate geometry

Puzzle Finding coordinates for given segment lengths

Example 3

a Plot the points $A(0, 6)$, $B(5, 6)$, $C(5, 2)$ and $D(-4, 2)$ on a number plane and join them to make the quadrilateral $ABCD$.

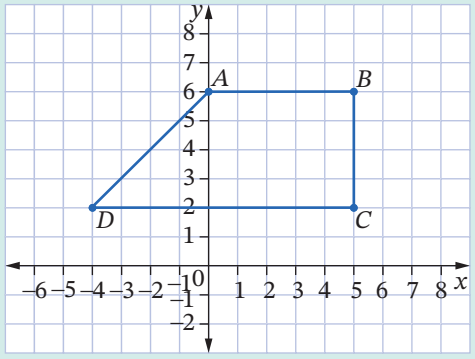
b What type of quadrilateral is $ABCD$?

c Find the exact length of AD.

d Hence find the perimeter of $ABCD$, correct to 2 decimal places.

Solution

a



Join the points in the correct order

b Since $AB \parallel CD$, the quadrilateral is a trapezium.

c $AD^2 = 4^2 + 4^2$

$= 32$

$AD = \sqrt{32}$

$= 4\sqrt{2}$ units

In exact surd form

d By counting grid squares, $AB = 5$, $BC = 4$, $CD = 9$.

Perimeter of $ABCD = 5 + 4 + 9 + 4\sqrt{2}$

$= 23.656...$

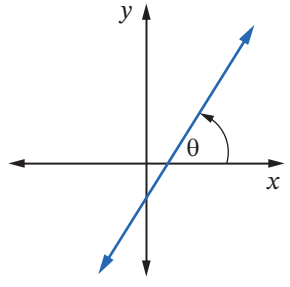
≈ 23.66 units

PATH

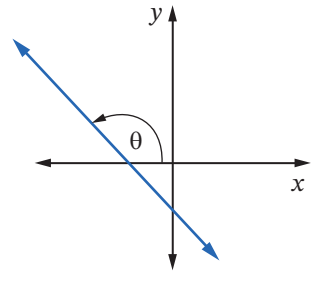
The angle of inclination of a line

The **angle of inclination**, θ, of a line is the angle it makes with the x-axis in the positive direction.

acute angle = positive gradient



obtuse angle = negative gradient

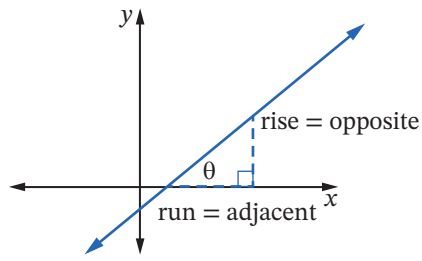


Note from the above diagrams that θ is **acute** when the line has a **positive gradient**, and **obtuse** when the line has a **negative gradient**.

We can use trigonometry to calculate the angle of inclination of a line using its gradient, m.

The diagram below shows that $m = \frac{\text{rise}}{\text{run}}$, but in trigonometry, $\tan\theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{\text{rise}}{\text{run}}$.

$\therefore m = \tan\theta$.



The angle of inclination of a line

The angle of inclination, θ, of a line is related to the gradient, m, of the line by the formula:

$$m = \tan\theta.$$

Example 4

Find, correct to the nearest degree, the angle of inclination of a line with gradient:

a $\frac{1}{3}$ **b** -4

Solution

a $m = \tan\theta$

$\frac{1}{3} = \tan\theta$

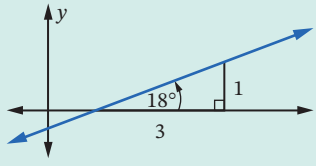
$\tan\theta = \frac{1}{3}$

$\theta = 18.4349...$

$\approx 18°$

The positive gradient means it is an acute angle.

On a calculator: SHIFT tan 1 $a^{b/c}$ 3 =



b $m = \tan\theta$

$-4 = \tan\theta$

$\tan\theta = -4$

$\theta = -75.9637...$

$\approx -76°$

But this negative angle is the angle below the x-axis.

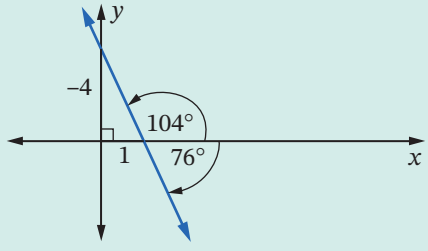
To find the angle of inclination,

$\theta \approx 180° - 76°$

$\approx 104°$

Negative gradient means it is an obtuse angle.

On calculator: SHIFT tan (−) 4 =

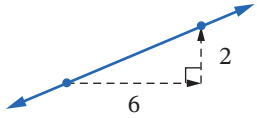


EXERCISE 2.01 ANSWERS ON P. 609

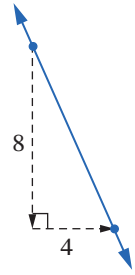
Length, midpoint and gradient of an interval U F PS R C

1 Calculate the gradient of each line.

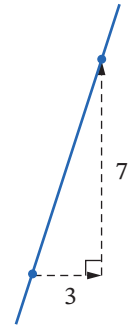
a



b



c



EXAMPLES 1, 2

2 For the interval joining each pair of points given, find:

i the length of the interval, correct to one decimal place

ii the midpoint of the interval

iii the gradient of the interval.

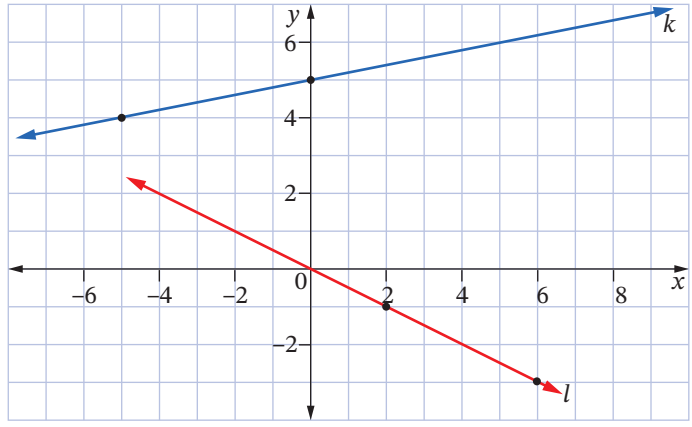
a $A(5, 3)$ and $B(7, 2)$ **b** $J(-1, 0)$ and $K(8, 6)$ **c** $M(0, -3)$ and $N(-5, 2)$

d $R(-3, -6)$ and $S(4, -9)$ **e** $A(-7, 2)$ and $B(-5, -8)$ **f** $U(3, -2)$ and $V(7, 2)$

3 Calculate, in exact (surd) form, the distance between each pair of points.

a $(-8, -1)$ and $(0, 4)$ **b** $(12, -6)$ and $(-1, -1)$ **c** $(7, -2)$ and $(-2, -3)$

4 Find the gradient of the lines labelled k and l.



5 Which expression gives the y-coordinate of the midpoint of the interval joining points $(3, 8)$ and $(-1, 5)$? Select **A**, **B**, **C** or **D**.

A $\frac{-1+5}{2}$ **B** $\frac{8+5}{2}$ **C** $\frac{8-5}{2}$ **D** $\frac{5-8}{2}$

Foundation Standard Complex

9780170479660

EXAMPLE 3

2.01

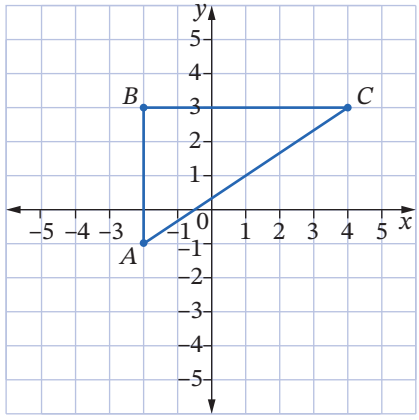
6 The vertices of triangle ABC are $A(-1, -1)$, $B(1, 3)$ and $C(3, 1)$. PS R C

- **a** Draw $\triangle ABC$ on a number plane.
- **b** Find the exact length of each side of the triangle.
- **c** Are any sides of the triangle equal in length?
- **d** What type of triangle is ABC?
- **e** Find the perimeter of $\triangle ABC$, correct to one decimal place.

7 The vertices of quadrilateral $KLMP$ are $K(1, 6)$, $L(7, 2)$, $M(3, -4)$ and $P(-3, 0)$. PS R C

- **a** Draw the quadrilateral on a number plane.
- **b** What type of quadrilateral is $KLMP$?
- **c** Find the gradients of sides KL and PM.
- **d** Find the gradients of sides KP and LM.
- **e** What do you notice about the gradients of the opposite sides of this quadrilateral? What does that mean about those sides?
- **f** Find the exact length of each side of $KLMP$.
- **g** Find the perimeter of $KLMP$, correct to one decimal place.
- **h** Find the area of $KLMP$.

8 This diagram shows a right-angled triangle with vertices $A(-2, -1)$, $B(-2, 3)$ and $C(4, 3)$. R C



- **a** Copy the diagram and find the coordinates of P and Q, the midpoints of BA and BC respectively. Mark P and Q on your diagram.
- **b** Calculate, correct to one decimal place, the lengths of PQ and AC. What do you notice about your answers?
- **c** Find the gradients of PQ and AC. What do you notice about your answers?

PATH

EXAMPLE 4

9 Find, correct to the nearest degree, the angle of inclination of a line with gradient:

a 3 **b** $\frac{1}{2}$ **c** 1 **d** 2.5

e -2 **f** $\frac{3}{4}$ **g** $-\frac{1}{10}$ **h** $-\frac{2}{3}$

10 Find, correct to 2 decimal places, the gradient of the line with angle of inclination:

a 60° **a** 158° **c** 42° **d** 94°

e 8° **f** 135° **g** 177° **h** 0°

Foundation Standard Complex

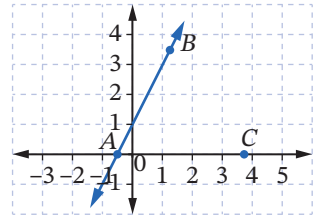
Technology

The angle of inclination

In this activity, we will use dynamic geometry software to calculate the angle of inclination of a line.

1 Use the software to graph the linear equation $y = 2x + 1$.

2 Create points A, B and C as shown and then measure the angle $\angle BAC$.



3 What is the angle of inclination of the line? Answer to the nearest degree.

4 Use the software to measure the angle of inclination of the line with equation:

a $y = -3x - 5$ **b** $y = x + 2$ **c** $y = x - 6$ **d** $y = -2x + 3$

e $y = -5x - 7$ **f** $y = -8x + 1$ **g** $y = 3x - 12$ **h** $y = \frac{2}{3}x + 4$

Investigation

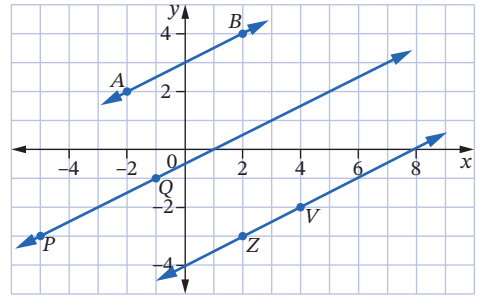
Parallel and perpendicular lines

1 These 3 lines are parallel. Calculate the gradient of:

a AB

b PQ

c ZV

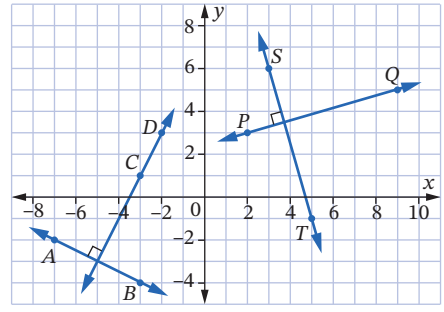


2 What can you conclude about the gradients of parallel lines?

3 This diagram shows 2 pairs of perpendicular lines. $AB \perp CD$ and $PQ \perp ST$. Calculate the gradient of:

a AB **b** CD

c PQ **d** ST



4 Is there a relationship between:

a the gradients of AB and CD? **b** the gradients of PQ and ST?

5 Calculate the product of:

a the gradients of AB and CD **b** the gradients of PQ and ST

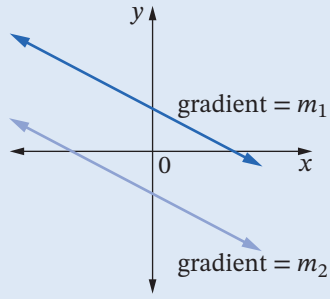
6 What can you conclude about the gradients of perpendicular lines?

Parallel and perpendicular lines 2.02

Parallel lines

Parallel lines have the same gradient.

If 2 lines with gradients m_1 and m_2 are parallel, then $m_1 = m_2$.



Puzzle
Gradients of parallel and perpendicular lines

2.02

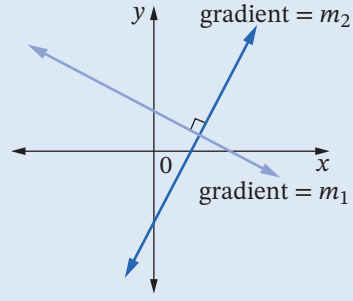
Perpendicular lines

Perpendicular lines have gradients with a product of -1.

If 2 lines with gradients m_1 and m_2 are perpendicular, then

$$m_1 \times m_2 = -1 \text{ or } m_2 = -\frac{1}{m_1}.$$

Note that m_2 is the **negative reciprocal** of m_1.



Example 5

State whether each pair of gradients represent parallel lines, perpendicular lines or neither.

a $m_1 = \frac{1}{2}, m_2 = 2$ **b** $m_1 = 0.4, m_2 = \frac{2}{5}$ **c** $m_1 = 1\frac{3}{5}, m_2 = -\frac{5}{8}$

Solution

a $m_1 \neq m_2$ so the lines are not parallel.

$$m_1 \times m_2 = \frac{1}{2} \times 2$$
$$= 1$$
$$\neq -1$$

so the lines are not perpendicular.

$\therefore$ The lines are neither parallel nor perpendicular.

b $m_2 = \frac{2}{5} = 0.4$

$m_1 = m_2$

$\therefore$ The lines are parallel.

c $m_1 = 1\frac{3}{5} = \frac{8}{5}$

$$m_1 \times m_2 = \frac{8}{5} \times \left(-\frac{5}{8}\right)$$
$$= -1$$

$\therefore$ The lines are perpendicular.

Video
Gradients of perpendicular lines

Example 6

Find the gradient of a line that is perpendicular to a line with gradient:

a 2 **b** -3 **c** $\frac{3}{4}$ **d** -0.6

Solution

a $m_1 = 2$

$m_2 = \frac{-1}{m_1}$ for perpendicular lines

$= \frac{-1}{2}$ The negative reciprocal of m_1.

$= -\frac{1}{2}$

The gradient is $-\frac{1}{2}$.

b $m_1 = -3$

$m_2 = \frac{-1}{m_1}$

$= \frac{-1}{-3}$

$= \frac{1}{3}$

The gradient is $\frac{1}{3}$.

c $m_1 = \frac{3}{4}$

$m_2 = \frac{-1}{m_1}$

$= \frac{-1}{\left(\frac{3}{4}\right)}$

$= -\frac{4}{3}$

The gradient is $-\frac{4}{3}$.

d $m_1 = -0.6 = -\frac{3}{5}$

$m_2 = \frac{-1}{\left(-\frac{3}{5}\right)} = \frac{5}{3}$

The gradient is $\frac{5}{3}$.

Example 7

A line passes through the points $A(-2, 5)$ and $B(4, 1)$. What is the gradient of a line:

a parallel to AB? **b** perpendicular to AB?

Solution

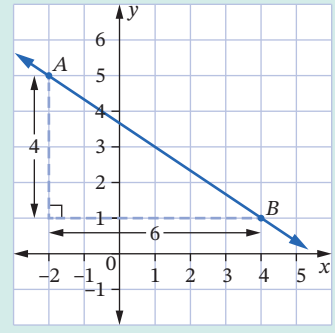
Find the gradient of AB by calculating the rise and run.

Sometimes drawing a diagram helps you to understand the problem.

Rise $= 1 - 5 = -4$

Run $= 4 - (-2) = 6$

Gradient $AB = \frac{-4}{6} = -\frac{2}{3}$



a Any line parallel to AB will have the same gradient as AB. $\therefore m = -\frac{2}{3}$

b The gradient of a line perpendicular to AB will be given by: $m = \frac{-1}{\left(-\frac{2}{3}\right)} = \frac{3}{2}$

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EXERCISE 2.02 ANSWERS ON P. 610

Parallel and perpendicular lines U F

EXAMPLE 5

1 State whether each pair of gradients represent parallel lines, perpendicular lines or neither.

a $m_1 = \frac{1}{4}, m_2 = 4$ **b** $m_1 = 3, m_2 = -\frac{1}{3}$ **c** $m_1 = 0.5, m_2 = \frac{1}{2}$

d $m_1 = \frac{2}{7}, m_2 = \frac{7}{2}$ **e** $m_1 = \frac{3}{10}, m_2 = 0.3$ **f** $m_1 = 1\frac{1}{5}, m_2 = -\frac{6}{5}$

2.02

2 Find the gradient of a line that is parallel to a line with gradient:

a 4 **b** -2 **c** $\frac{1}{3}$ **d** -0.2

EXAMPLE 6

3 Find the gradient of a line that is perpendicular to a line with gradient:

a 1 **b** -6 **c** -1.5 **d** $\frac{5}{2}$

4 What is the gradient of a line that is perpendicular to a line with a gradient of 0.8? Select the correct answer **A**, **B**, **C** or **D**.

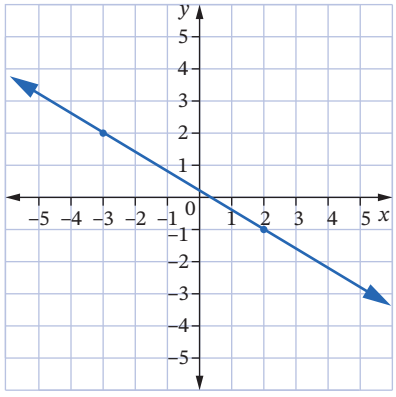
A 0.2 **B** -0.2 **C** 1.25 **D** -1.25

5 What is the gradient of a line that is parallel to a line that goes through $P(0, 3)$ and $Q(5, -2)$? Select **A**, **B**, **C** or **D**.

A 1 **B** -1 **C** $\frac{1}{5}$ **D** $-\frac{1}{5}$

EXAMPLE 7

6 What is the gradient of a line perpendicular to the line shown? Select **A**, **B**, **C** or **D**.



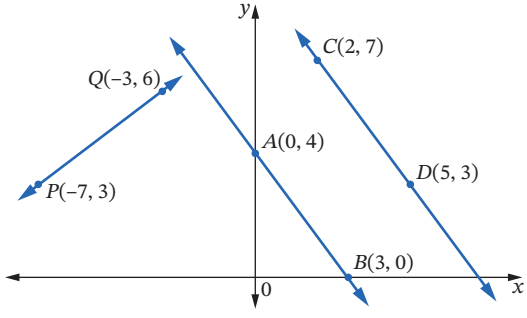
A $\frac{5}{3}$ **B** -5

C $\frac{3}{5}$ **D** $\frac{1}{5}$

7 Calculate the gradient of each line shown and test whether:

a $AB \parallel CD$

b $PQ \perp CD$.



8 A line passes through the points $R(-5, 2)$ and $S(1, 4)$. What is the gradient of a line:

a parallel to RS? **b** perpendicular to RS?

Foundation Standard Complex

Technology

Parallel and perpendicular lines

Use dynamic geometry software to find out if sets of linear equations represent parallel or perpendicular lines.

1 Parallel lines

Graph the following lines and use the **Slope/Gradient** function to find their gradients and check whether they are parallel using $m_1 = m_2$.

a $5x - 3y = 0$ and $y = \frac{5x}{3}$

b $x + y + 4 = 0$ and $x + y - 6 = 0$

c $x - 2y = 0$ and $y = 0.5x$

d $y = 5x - 9$ and $5x - y - 1 = 0$

2 Perpendicular lines

Graph the following lines and use the **Slope/Gradient** function to find their gradients and check whether they are perpendicular using $m_1 \times m_2 = -1$.

a $y = 0.6x + 2$ and $y = \frac{5}{3}x$

b $x - 4y + 1 = 0$ and $y = -4x - 3$

c $3x - 2y = 0$ and $y = -\frac{2x}{3}$

d $y = 2x + 4$ and $x - 2y - 1 = 0$

2.03 The gradient-intercept equation $y = mx + c$

PATH

Video
Graphing linear equations

Interactives
Graphing lines
Function builder: Basics
Function builder

Worksheets
A page of lines
A page of number planes
Graphing linear equations
Graphing linear equations (Advanced)

Skillsheet
Graphing linear equations

Using *x*- and *y*-intercepts to graph lines

- To find the *x*-intercept, substitute $y = 0$ and solve the equation.
- To find the *y*-intercept, substitute $x = 0$ and solve the equation.

Example 8

Find the *x*- and *y*-intercepts of the line $2x - 3y = 6$ to draw its graph.

Solution

For *x*-intercept, $y = 0$:

$2x - 3 \times 0 = 6$

$2x = 6$

$x = 3$

The *x*-intercept is 3.

For *y*-intercept, $x = 0$:

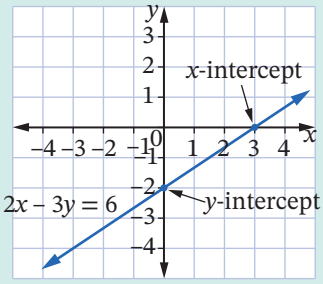
$2 \times 0 - 3y = 6$

$-3y = 6$

$y = -2$

The *y*-intercept is –2.

Plot both intercepts on the axes, draw a line through the 2 points and label the line with its equation.



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Testing if a point lies on a line

A point lies on a line if its (x, y) coordinates satisfy the equation of the line.

Video
Testing if a point lies on a line

Worksheet
Straight-line equations

2.03

Example 9

Test whether each point lies on the line $x - 2y = 5$.

a $(17, 6)$ **b** $(8, -4)$

Solution

- Separate the equation into its left-hand side (LHS) and right-hand side (RHS).
- Substitute the coordinates of the point into both sides.
- If LHS = RHS, the point satisfies the equation and so lies on the line.
- If LHS $\neq$ RHS, the point does not lie on the line.

a Substitute $x = 17, y = 6$ into $x - 2y = 5$.

$\text{LHS} = x - 2y$

$= 17 - 2 \times 6$

$= 5$

$\text{RHS} = 5$

LHS = RHS, so $(17, 6)$ lies on the line.

b Substitute $x = 8, y = -4$ into $x - 2y = 5$.

$\text{LHS} = x - 2y$

$= 8 - 2 \times (-4)$

$= 16$

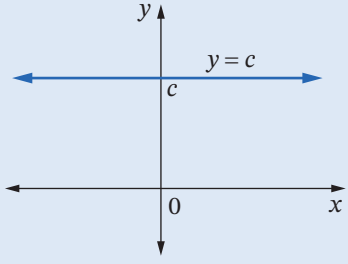
$\text{RHS} = 5$

LHS $\neq$ RHS, so $(8, -4)$ does not lie on the line.

Horizontal and vertical lines

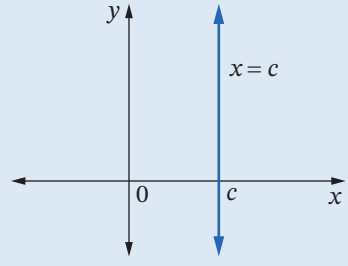
Video
Horizontal and vertical lines

The **equation of a horizontal line** is of the form $y = c$ (where c is a constant number).



Note: The x-axis has equation $y = 0$.

The **equation of a vertical line** is of the form $x = c$ (where c is a constant number).



Note: The y-axis has equation $x = 0$.

Example 10

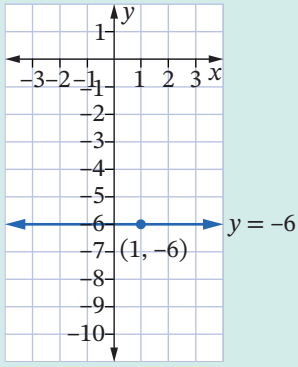
Find the equation of the line that is:

a parallel to the x-axis and passes through the point $(1, -6)$

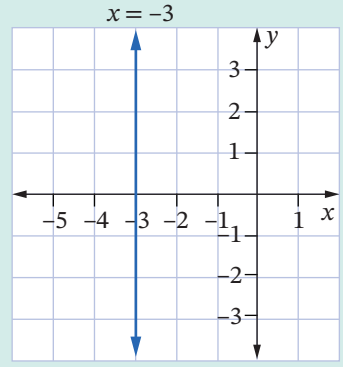
b always 3 units to the left of the y-axis.

Solution

a The equation of the line is $y = -6$.



b The equation of the line is $x = -3$.



The gradient–intercept form of a linear equation

The equation of a straight line is $y = mx + c$, where m is the **gradient** and c is the ***y*-intercept**.

For this reason, $y = mx + c$ is also called the **gradient–intercept form** of a linear equation.

Videos
Graphing $y = mx + c$

Gradient and y-intercept of a line

The gradient-intercept formula

Example 11

Find the gradient and y-intercept of the line with equation:

a $y = -4x + 9$ **b** $y = 10 - 6x$ **c** $y = \frac{5x+4}{2}$ **d** $3x + 2y - 6 = 0$

Solution

a $y = -4x + 9$ is in the form $y = mx + c$. $\therefore$ Gradient $m = -4$ and y-intercept $c = 9$.

b $y = 10 - 6x$ can be rewritten as $y = -6x + 10$. $\therefore$ Gradient -6 and y-intercept 10.

c $y = \frac{5x+4}{2}$ can be rewritten as $y = \frac{5x}{2} + \frac{4}{2} = \frac{5x}{2} + 2$. $\therefore$ Gradient $\frac{5}{2}$ and y-intercept 2.

d $3x + 2y - 6 = 0$ can be rearranged in the form $y = mx + c$.

Make y the **subject** of the equation: 'solve' the equation for y.

$$2y - 6 = -3x$$
$$2y = -3x + 6$$
$$\frac{2y}{2} = \frac{-3x+6}{2}$$
$$y = \frac{-3x}{2} + 3$$

$\therefore$ Gradient $m = -\frac{3}{2}$ and y-intercept $c = 3$.

Interactive
Graphing slope-intercept

Puzzles
Equation of a line

Equations in gradient form

Spreadsheet
Drawing straight lines: $y = mx + c$

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Example 12

Graph each linear equation by finding the gradient and y-intercept first.

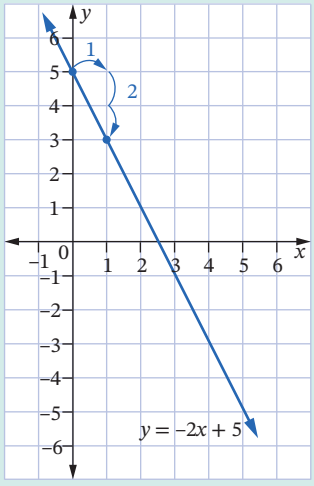
a $y = -2x + 5$ **b** $y = \frac{3}{4}x - 2$

Solution

a $y = -2x + 5$ has a gradient of -2 and a y-intercept of 5.

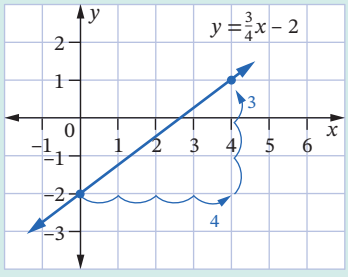
- Plot the y-intercept 5 on the y-axis.
- Make a gradient of $-2 = \frac{-2}{1}$ by moving **across** 1 unit (**run**) and **down** 2 units (**'negative' rise**) and marking the point at (1, 3).
- Rule a line through this point and the y-intercept.

Don't forget to label the line with its equation '$y = -2x + 5$'



b $y = \frac{3}{4}x - 2$ has gradient $\frac{3}{4}$ and y-intercept -2.

- Plot the y-intercept -2 on the y-axis.
- Make a gradient of $\frac{3}{4}$ by moving **across** 4 units (**run**) and **up** 3 units (**rise**) and marking the point at (4, 1).
- Rule a line through this point and the y-intercept.



Example 13

Test whether each line is parallel to $y = -2x + 3$.

a $y = 2x + 3$ **b** $y = -2x + 1$ **c** $y = -2x$ **d** $y = 5x + 3$

Solution

Parallel lines have the **same gradient**. The line $y = -2x + 3$ has the gradient $m = -2$.

a $y = 2x + 3$ has a different gradient, 2, so it is not parallel to $y = -2x + 3$.

b $y = -2x + 1$ has the same gradient, -2, so it is parallel to $y = -2x + 3$.

c $y = -2x$ has the same gradient, -2, so it is parallel to $y = -2x + 3$.

d $y = 5x + 3$ has a different gradient, 5, so it is not parallel to $y = -2x + 3$.

$\therefore$ The lines **b** $y = -2x + 1$ and **c** $y = -2x$ are parallel to $y = -2x + 3$.

EXERCISE 2.03 ANSWERS ON P. 610

The gradient-intercept equation $y = mx + c$ UFRC

PATH

EXAMPLE 8

1 Graph each linear equation after finding its x- and y-intercepts.

a	$y = 4 - 2x$	**b**	$2x = 4y - 8$	**c**	$y - x = 6$
d	$3x - 2y = 12$	**e**	$2x + 2y = 5$	**f**	$6 - x = 2y$
g	$y = 4 + 2x$	**h**	$5x + 3y - 15 = 0$	**i**	$3x - y = 6$
j	$2x - 5y - 20 = 0$	**k**	$4x + 2y - 8 = 0$	**l**	$x - 4y - 2 = 0$

EXAMPLE 9

2 Test whether the point $(3, -1)$ lies on each line.

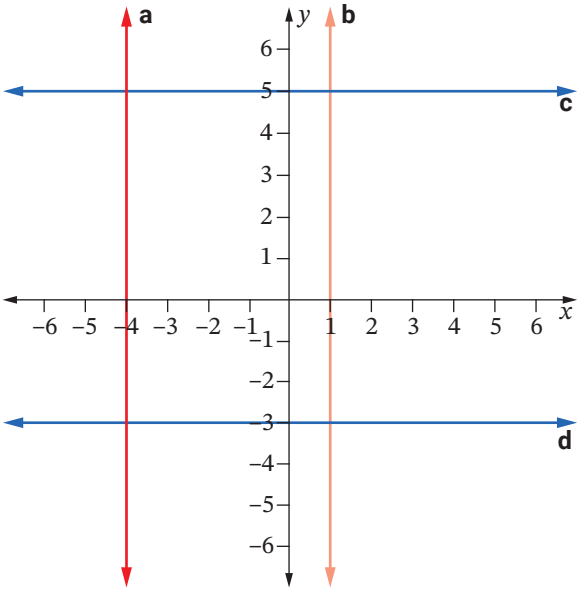
a	$y = 2x - 5$	**b**	$x - y = 4$	**c**	$y + 2x = 5$
d	$y = x - 4$	**e**	$x + y = 5$	**f**	$3x + y + 8 = 0$

3 Which point lies on the line $y = 6x - 5$? Select the correct answer **A**, **B**, **C** or **D**.

A $(-1, 11)$ **B** $(3, -13)$ **C** $(-2, -17)$ **D** $(-5, 25)$

EXAMPLE 10

4 Find the equation of each line shown.



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5 Find the equation of the line that is: R C

a horizontal and passes through the y-axis at 2

b vertical with an x-intercept of 4

c parallel to the y-axis and passes through the point $(-1, 4)$

d parallel to the x-axis and passes through the point $(0, -2)$

e 3 units above the x-axis

f 1 unit to the left of the y-axis

g drawn through the points $(-1, 6)$ and $(2, 6)$

h drawn through the points $(-1, 8)$ and $(-1, 2)$.

6 **a** What is another name for the line $y = 0$? C

b What is another name for the line $x = 0$? C

EXAMPLE 11

7 Find the gradient and y-intercept of each line.

a $y = 3x - 2$ **b** $y = -2x + 7$ **c** $y = x + 4$

d $y = 9 - x$ **e** $y = \frac{3x}{4} + 6$ **f** $y = x$

g $y = \frac{x}{2} - 11$ **h** $y = \frac{2x + 18}{3}$ **i** $y = \frac{-24 - x}{3}$

j $y = 2(x - 3)$ **k** $11 - 3x = y$ **l** $\frac{2x - 7}{2} = y$

EXAMPLE 12

8 Graph each linear equation by finding the gradient and y-intercept first.

a $y = 2x + 1$ **b** $y = 3x - 2$ **c** $y = 2x$ **d** $y = \frac{x}{2} - 1$

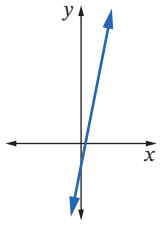
e $y = -2x + 3$ **f** $y = -\frac{3x}{4}$ **g** $y = \frac{-5x + 2}{2}$ **h** $y = \frac{3x - 20}{5}$

9 Write the equation of a line with a gradient of 2 that passes through the origin. R

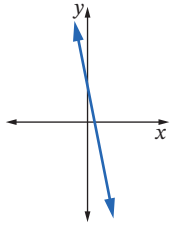
10 Match each equation to its graph below. R

a $y = 4x + 1$ **b** $y = -4x + 1$ **c** $y = -4x - 1$ **d** $y = 4x - 1$

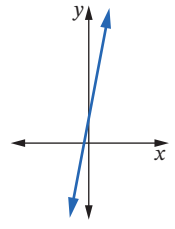
A



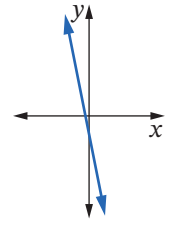
B



C



D



Foundation Standard Complex

EXAMPLE 13

11 For each given line, select the lines that are parallel to it. There may be more than one answer. R

a $y = x + 6$

A $y = 6x$ **B** $y = 6 - x$ **C** $y = x + 1$ **D** $y = 2x$

b $y = 3x + 10$

A $y = 10x + 3$ **B** $y = 3x - 1$ **C** $y = 1 - 3x$ **D** $y = 4 + 3x$

c $y = \frac{x}{2} + 5$

A $y = 2x - 1$ **B** $y = \frac{x+6}{2}$ **C** $y = 1 - \frac{x}{2}$ **D** $y = x + 2$

d $y = 6$

A $y = 2x + 6$ **B** $y = 6x$ **C** $y = -1$ **D** $y = 10$

e $y = 4x$

A $y = 4x - 2$ **B** $y = 4x + 3$ **C** $y = 4$ **D** $y = 1 - 4x$

f $x = 10$

A $y = 10$ **B** $y = 10x$ **C** $x = 2y$ **D** $x = -6$

12 For each set of linear equations, find a pair of equations whose graphs are parallel lines. R

a $y = 4x + 3$ $y = x + 2$ $y = 4x - 6$ $y = 2x$

b $y = 5x + 1$ $3x - y + 7 = 0$ $y = 3x - 2$ $y = -5x + 2$

Quiz
Mental skills 2

Mental skills 2: Maths without calculators ANSWERS ON P. 611

Finding 15%, $2\frac{1}{2}$%, 25% and $12\frac{1}{2}$%

- To find 10% or $\frac{1}{10}$ of a number, divide by 10.
- To find 5% of a number, find 10% first, then halve it (since 5% is half of 10%).
- So to find 15% of a number, find 10% and 5% of the number separately, then add the answers together.

1 Study each example.

a $15\% \times 3600 = (10\% \times 3600) + (5\% \times 3600) = 360 + 180 = 540$

b $15\% \times \$28 = (10\% \times \$28) + (5\% \times \$28) = \$2.80 + \$1.40 = \$4.20$

2 Now find 15% of each amount.

a	120	**b**	\$840	**c**	260	**d**	\$202
e	\$50	**f**	72	**g**	\$180	**h**	400
i	\$1600	**j**	\$22	**k**	6000	**l**	\$350

To find $2\frac{1}{2}$% of a number, first find 5%, then halve it.

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3 Study each example.

a $2\frac{1}{2}\% \times 600$

$10\% \times 600 = 60$

$5\% \times 600 = \frac{1}{2} \times 60 = 30$

$2\frac{1}{2}\% \times 600 = \frac{1}{2} \times 30 = 15$

b $2\frac{1}{2}\% \times \$820$

$10\% \times \$820 = \$82$

$5\% \times \$820 = \frac{1}{2} \times 82 = \$41$

$2\frac{1}{2}\% \times \$820 = \frac{1}{2} \times \$41 = \$20.50$

4 Now find $2\frac{1}{2}\%$ of each amount

a	400	**b**	6640	**c**	\$2000	**d**	\$880
e	1500	**f**	\$232	**g**	5400	**h**	\$904

To find 25% of a number, halve the number twice as $25\% = \frac{1}{4}$.

5 Study each example.

a $25\% \times 700$

$50\% \times 700 = \frac{1}{2} \times 700 = 350$

$25\% \times 700 = \frac{1}{2} \times 350 = 175$

b $25\% \times \$86$

$50\% \times \$86 = \frac{1}{2} \times \$86 = \$43$

$\therefore 25\% \times \$86 = \frac{1}{2} \times \$43 = \$21.50$

6 Now find 25% of each amount.

a	2000	**b**	\$80	**c**	18	**d**	\$25
e	\$324	**f**	\$140	**g**	66	**h**	298
i	\$780	**j**	\$1700	**k**	\$126	**l**	1160

To find $12\frac{1}{2}\%$ of a number, find 25% first, then halve it. In other words, halve 3 times.

7 Study each example.

a $12\frac{1}{2}\% \times 400$

$50\% \times 400 = \frac{1}{2} \times 400 = 200$

$25\% \times 400 = \frac{1}{2} \times 200 = 100$

$12\frac{1}{2}\% \times 400 = \frac{1}{2} \times 100 = 50$

b $12\frac{1}{2}\% \times \$144$

$50\% \times \$144 = \frac{1}{2} \times \$144 = \$72$

$25\% \times \$144 = \frac{1}{2} \times \$72 = \$36$

$12\frac{1}{2}\% \times \$144 = \frac{1}{2} \times \$36 = \$18$

8 Now find $12\frac{1}{2}\%$ of each amount.

a	1280	**b**	\$12	**c**	60	**d**	\$260
e	\$540	**f**	\$250	**g**	304	**h**	1360

2.04 The general equation $ax + by + c = 0$

PATH

Worksheet
Parallel and perpendicular lines

Puzzle
Linear equations code puzzle

Video
The general equation $ax + by + c = 0$

A linear equation written in **gradient–intercept form**, such as $y = -\frac{3}{4}x + 2$, can also be written in **general form** as $3x + 4y - 8 = 0$. Note that, for the general form, all of the terms on the left-hand side of the equation are written with no fractions, and only 0 is on the right-hand side. Sometimes the general form is neater and more convenient.

The general form of a linear equation

The **general form of a linear equation** is written as $ax + by + c = 0$, where a, b and c are integers and a is positive.

Example 14

Write each linear equation in general form.

a $y = 6x + 2$ **b** $y = -\frac{2}{3}x + 2$ **c** $y = 2x - \frac{3}{5}$

Solution

a $y = 6x + 2$

$0 = 6x - y + 2$ Subtracting y from both sides

$6x - y + 2 = 0$ Swapping sides so that 0 appears on the RHS

b $y = -\frac{2}{3}x + 2$

$3y = 3\left(-\frac{2}{3}x + 2\right)$ Multiplying both sides by 3 to remove the fraction

$3y = -2x + 6$

$2x + 3y = 6$ Adding $2x$ to both sides

$2x + 3y - 6 = 0$ Subtracting 6 from both sides

c $y = 2x - \frac{3}{5}$

$5y = 5\left(2x - \frac{3}{5}\right)$ Multiplying both sides by 5

$5y = 10x - 3$

$0 = 10x - 5y - 3$ Subtracting $5y$ from both sides

$10x - 5y - 3 = 0$ Swapping sides so that 0 appears on the RHS

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Example 15

Find the gradient and y-intercept of the line whose equation is $5x + 2y - 10 = 0$.

Solution

Rewrite $5x + 2y - 10$ in the form $y = mx + c$.

Make y the subject, solve for y.

$5x + 2y - 10 = 0$

$2y - 10 = -5x$ Subtracting $5x$ from both sides

$2y = -5x + 10$ Adding 10 to both sides

$y = \frac{-5x + 10}{2}$ Dividing both sides by 2

Aim to have y on its own on the LHS of the equation.

$y = \frac{-5x}{2} + 5$

$\therefore$ Gradient: $m = -\frac{5}{2}$

y-intercept: $c = 5$

Example 16

Find the x- and y-intercepts of each line.

a $3x - 2y + 12 = 0$ **b** $x + 3y - 7 = 0$

Solution

a For x-intercept, substitute $y = 0$:

$3x - 2(0) + 12 = 0$

$3x = -12$

$x = -4$

For y-intercept, substitute $x = 0$:

$3(0) - 2y + 12 = 0$

$-2y = -12$

$y = 6$

b For x-intercept, substitute $y = 0$:

$x + 3(0) - 7 = 0$

$x = 7$

For y-intercept, substitute $x = 0$:

$0 + 3y - 7 = 0$

$3y = 7$

$y = \frac{7}{3}$

EXERCISE 2.04 ANSWERS ON P. 611

The general equation $ax + by + c = 0$ U F R C

1 Write each linear equation in general form. R C

EXAMPLE 14

a $y = x + 2$ **b** $y = 3x - 1$ **c** $y = 8 + 5x$

d $x + 2y = 3$ **e** $x - 2y = 6$ **f** $y = 8x + 2$

g $y + 3 = 6x$ **h** $2y = x - 6$ **i** $y = \frac{3}{5}x + 2$



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EXAMPLE 15

2 Find the gradient and y-intercept of the line with equation: R C

a $2x + y = 6$ **b** $8x - 2y = 10$ **c** $3x - 2y + 4 = 0$

d $y + 2x - 1 = 0$ **e** $2x + y + 5 = 0$ **f** $4x + 3y - 12 = 0$

3 Find the gradient, m, and the y-intercept, c, of the line with equation $x - 3y + 5 = 0$. Select the correct answer **A**, **B**, **C** or **D**.

A $m = -1, c = 5$ **B** $m = \frac{1}{3}, c = \frac{5}{3}$ **C** $m = 1, c = -5$ **D** $m = \frac{1}{3}, c = -\frac{5}{3}$

EXAMPLE 16

4 Find the x- and y-intercepts of each line in Question **2**.

5 Which statement is FALSE about the line whose equation is $3x + y - 6 = 0$? Select **A**, **B**, **C** or **D**. R C

A Its gradient is -3. **B** Its y-intercept is -6.

C Its x-intercept is 2. **D** It is parallel to the line $y = -3x$.

6 Express each linear equation in general form. R C

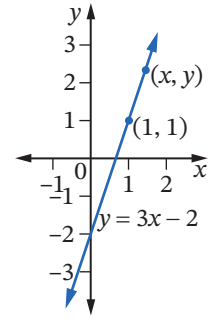
a $y = \frac{x}{3} + \frac{1}{4}$ **b** $y = -\frac{5x}{8} + \frac{3}{2}$ **c** $y = \frac{3x}{4} - \frac{2}{5}$

d $y = -\frac{x}{2} - \frac{3}{4}$ **e** $y = \frac{3x}{2} + \frac{2}{5}$ **f** $y = -\frac{9x}{10} - \frac{7}{15}$

Investigation

A line, its gradient and a point

1 The graph shows the line $y = 3x - 2$.



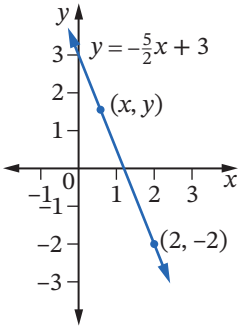
a What is its gradient?

b If (x, y) is any other point on the line, show that $m = \frac{y-1}{x-1}$.

c Explain why $\frac{y-1}{x-1} = 3$.

d Hence show that this equation simplifies to $y = 3x - 2$.

2 The graph shows the line $y = -\frac{5}{2}x + 3$.



a What is its gradient?

b If (x, y) is any other point on the line, show that $m = \frac{y+2}{x-2}$.

c Explain why $\frac{y+2}{x-2} = -\frac{5}{2}$

d Hence show that this equation simplifies to $y = -\frac{5}{2}x + 3$.

3 The equation of a line is given by $\frac{y-2}{x-7} = \frac{3}{4}$.

a What is the gradient of the line?

b Can you give the coordinates of a point on this line by looking at its equation? Why?

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4 Write the equation of a line that passes through the point (−3, 5) and has a gradient equal to 2. Compare your result with other groups.

5 A line with gradient m passes through the point (x_1, y_1).

a Show that $m = \frac{y - y_1}{x - x_1}$, where (x, y) is any other point on the line.

b Explain why $y - y_1 = m(x - x_1)$.

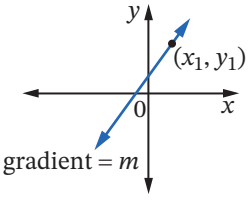
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The point–gradient equation $y - y_1 = m(x - x_1)$

2.05

PATH

There is a formula for finding the equation of a line if we know its gradient m and a point on the line (x_1, y_1). Let (x, y) be any other point on the line.



Then $m = \frac{y - y_1}{x - x_1}$

or $y - y_1 = m(x - x_1)$.

This is called the **point–gradient form** of a linear equation.

The point–gradient form of a linear equation

The equation of a line with gradient m and which passes through the point (x_1, y_1) is:

$$y - y_1 = m(x - x_1)$$

Example 17

Find the equation of the line with a gradient of $\frac{2}{3}$ that passes through the point (−2, 1).

Solution

$m = \frac{2}{3}, x_1 = -2, y_1 = 1.$

$y - y_1 = m(x - x_1)$

$y - 1 = \frac{2}{3}[x - (-2)]$

$3(y - 1) = 2(x + 2)$

$3y - 3 = 2x + 4$

$0 = 2x - 3y + 7$

$2x - 3y + 7 = 0$

In general form

Video
The point–gradient formula

Example 18

Find the equation of the line passing through the points (1, 3) and (4, −3).

Solution

First find the gradient of the line by using the points (1, 3) and (4, −3).

$m = \frac{-3 - 3}{4 - 1}$ (x_1, y_1) (x_2, y_2)

$= \frac{-6}{3}$

$= -2$

Video
The point-gradient formula

PATH

Now use $y - y_1 = m(x - x_1)$ with $m = -2$ and $(1, 3)$.

Either point (1, 3) or (4, –3) can be used to find the equation of the line.

$y - 3 = -2(x - 1)$

$= -2x + 2$

$y = -2x + 5$ or $2x + y - 5 = 0$ in general form

OR: using the other point (4, –3) instead:

$y - (-3) = -2(x - 4)$

$y + 3 = -2x + 8$

$y = -2x + 5$ or $2x + y - 5 = 0$ in general form

EXERCISE 2.05 ANSWERS ON P. 611

The point–gradient equation $y - y_1 = m(x - x_1)$ U F R C

In this exercise, express all equations of lines in general form.

EXAMPLE 17

1 Find the equation of each line, given a point on the line and the gradient.

a (2, 5), gradient 2 **b** (–6, 4), gradient –1 **c** (3, –8), gradient 4

d (–1, –2), gradient $\frac{2}{3}$ **e** (2, –8), gradient $-\frac{1}{5}$ **f** (–1, 7), gradient –3

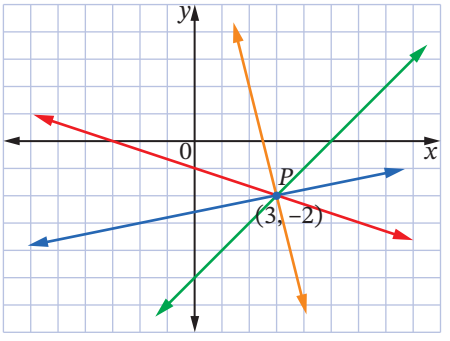
g $\left(\frac{1}{2}, -3\right)$, gradient –4 **h** $\left(-4, -\frac{1}{2}\right)$, gradient $\frac{3}{4}$ **i** (–2, –6), gradient –2

2 4 lines a, b, c and d intersect at $P(3, -2)$.

The gradients of a, b, c and d are $1, -\frac{1}{3}, -4$ and $\frac{1}{5}$ respectively. R C

a Copy the diagram and correctly label the lines a, b, c and d.

b Find the equation of each line.



EXAMPLE 18

3 Find the equation of the line passing through each pair of points.

a (7, 3) and (10, 6) **b** (8, 10) and (–2, 2) **c** (–1, 3) and (5, 8)

d (2, –2) and (–1, 6) **e** (4, –3) and (6, –6) **f** (–1, –2) and (2, 3)

g (–10, 2) and (1, –4) **h** (–3, 6) and (1, 2) **i** (–4, –9) and (–1, –5)

4 2 lines k and l intersect at (–1, 4). Line k has a gradient of $-\frac{1}{2}$, while line l has a gradient of 3. Find the equations of lines k and l. R C

5 Find the equation of a line with a gradient of –4 and an x-intercept of 5. R C

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6 A line passes through the y-axis at $(0, 6)$ and has a gradient of $\frac{5}{7}$. What is its equation? R C

7 A line with a gradient of $\frac{2}{3}$ passes through the midpoint of $(5, 6)$ and $(1, 10)$.

Find the equation of the line. R C

8 A line with a gradient of $-\frac{3}{5}$ passes through the midpoint of $(-8, -2)$ and $(-2, 20)$.

Find its equation. R C

9 The gradient–intercept form of a line, $y = mx + c$, can also be used to find the equation of a line, given its gradient and a point on the line. R C

a Use $y = mx + c$ to find the equation of the line with gradient 2 that passes through the point $(2, 5)$.

b Compare your equation with your answer to question **1a**.

10 The point–gradient formula can be converted to a formula for finding the equation of a line passing through 2 points (x_1, y_1) and (x_2, y_2). R C

a Prove that the 'two-point formula' is $\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$.

b Use the two-point form to find the equation of a line passing through the points $(7, 3)$ and $(10, 6)$.

c Compare your equation with your answer to question **3a**.

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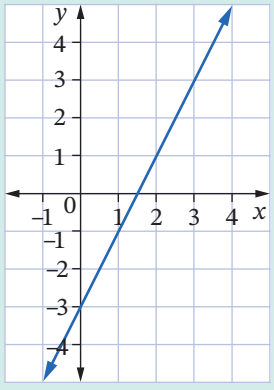
Finding the equation of a line

2.06

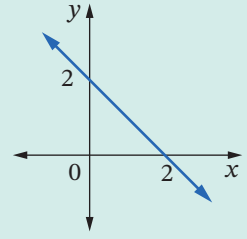
Example 19

Find the equation of each line.

a



b



Video
Finding the equation of a line

Worksheets
Finding the equation of a line

A page of lines

Puzzle
Straight-line equations

Solution

a Select 2 points on the line to find the gradient, say (0, −3) and (2, 1).

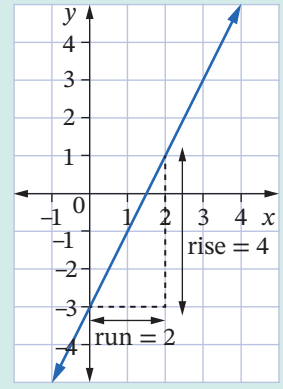
Gradient: $m = \frac{\text{rise}}{\text{run}} = \frac{4}{2} = 2$ from the graph

y-intercept: $c = -3$

∴ The equation of the line is $y = 2x - 3$. $y = mx + c$

We can check that this equation is correct for any point on the line, say (3, 3).

When $x = 3$, $y = 2 \times 3 - 3 = 3$.

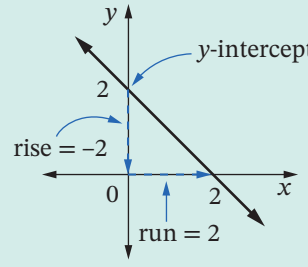


b Find the gradient of the line passing through (0, 2) and (2, 0).

Gradient: $m = \frac{\text{rise}}{\text{run}} = \frac{-2}{2} = -1$ from the graph

y-intercept: $c = 2$

∴ The equation of line is $y = -x + 2$. $y = mx + c$

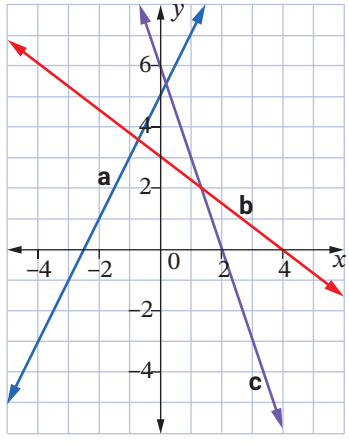


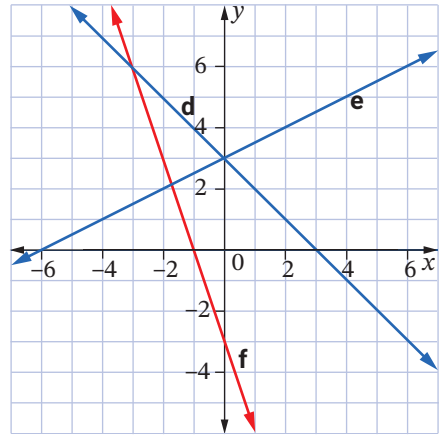
EXERCISE 2.06 ANSWERS ON P. 612

Finding the equation of a line U F R C

EXAMPLE 19

1 Find the equation of each line. R C



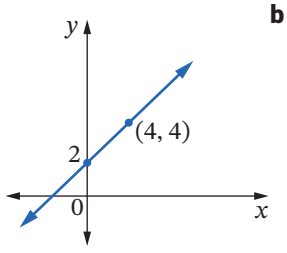


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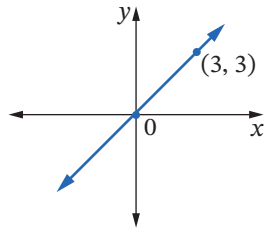
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2 Find the equation of each line. R C

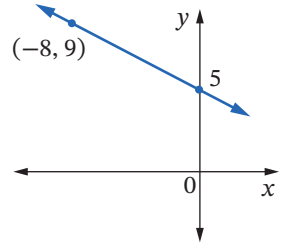
a



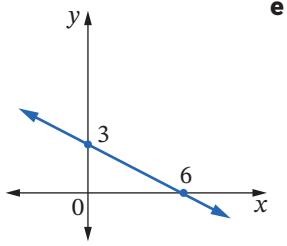
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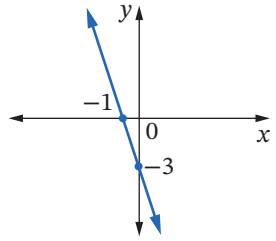
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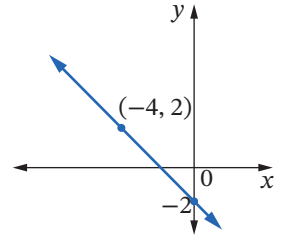
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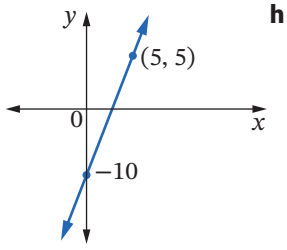
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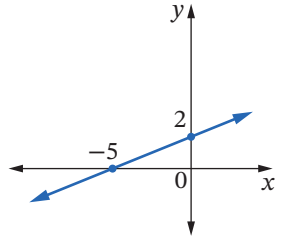
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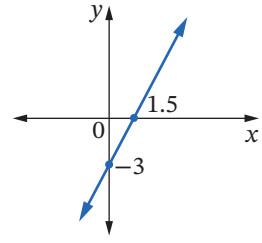
g



h



i



Applying linear equations 2.07

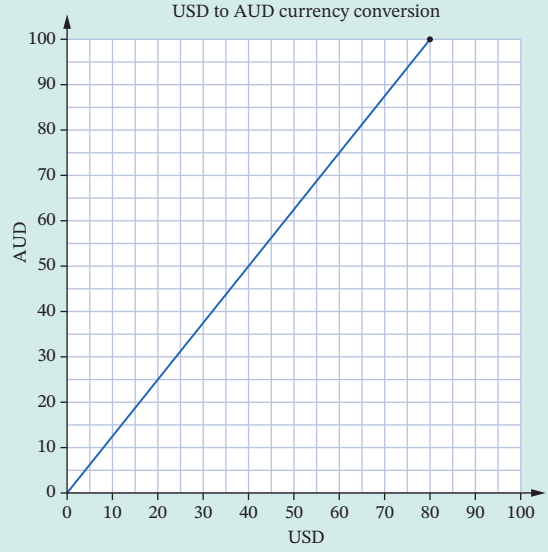
Linear equations can be used to model real-life situations such as the cost of a taxi trip or the exchange rate of currencies between 2 countries or the amount of water in a swimming pool when it is being drained. Instead of using x and y as our variables in the linear equation $y = mx + c$, we can use other letters of the alphabet to represent quantities.

Video Applying linear functions

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Example 20

This conversion graph shows the exchange rate between USD (US dollars) and AUD (Australian dollars)



a Find the gradient and vertical intercept of the line.

b What does the gradient represent?

c Write the equation of the line in the form $A = mU + c$ using U for USD and A for AUD.

d Use the graph to convert:

i 60 USD to AUD

ii 25 AUD to USD.

Solution

a Line goes from (0, 0) to (80, 100).

Gradient $= \frac{100-0}{80-0} = 1.25$

Vertical intercept $= 0$ (from graph)

Vertical intercept is the general name for the y-intercept when a variable other than y is used.

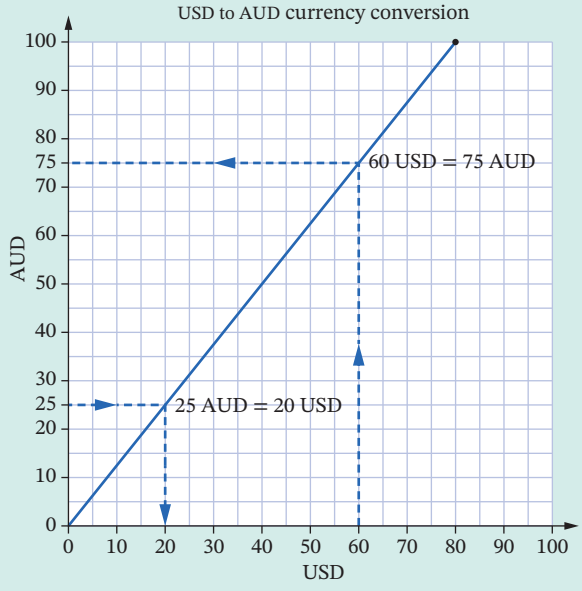
b The gradient, 1.25, represents the exchange rate between USD and AUD (1 USD = 1.25 AUD)

c $A = mU + c$ and we know that $m = 1.25$ and $c = 0$.

We're replacing x with U because U is on the horizontal (x) axis and y with A because A is on the vertical (y) axis.

The equation of the line is $A = 1.25U + 0$, which simplifies to $A = 1.25U$.

d Using the graph:



i 60 USD = 75 AUD

ii 25 AUD = 20 USD

Applying linear equations

- The **gradient** represents the **rate of increase or decrease** in the situation.
- For an increase, the gradient is positive and for a decrease, the gradient is negative.
- The *y*-**intercept** or **vertical intercept** represents the **starting value** in the situation.

EXERCISE 2.07 ANSWERS ON P. 612

Applying linear equations U F R C

These questions may also be completed using a spreadsheet or graphing software.

1 Write an equation in the form $y = mx + c$ using the given change of variables and values. C

a $y \to D, x \to h, m = 30, c = 5$

b $y \to R, x \to t, m = 0.5, c = -8$

c $y \to P, x \to n, m = -50, c = 0$

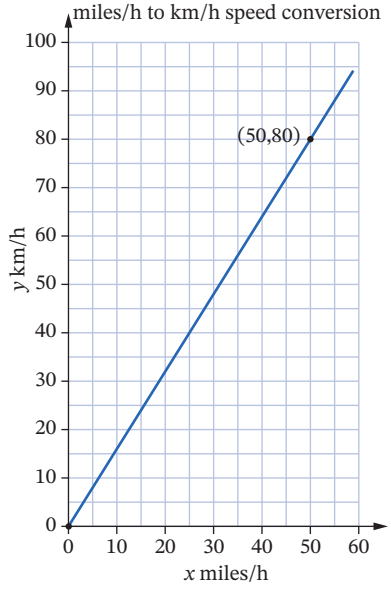
d $y \to F, x \to C, m = 1.8, c = 32$

e $y \to H, x \to w, m = 0.8, c = 0$

f $y \to V, x \to T, m = -40, c = 200$

EXAMPLE 20

2 This conversion graph converts between speeds in miles per hour to km/h. R C



a Use the graph to find what speed in km/h is equivalent to 25 miles/h.

b Find the gradient and vertical intercept of the line.

c What does the gradient represent?

d Find the equation of the line.

3 The distance d km travelled by a car moving at a constant speed of 80 km/h after t hours is given by the equation $d = 80t$. R C

a Copy and complete this table of values using the equation, then graph the equation on a number plane.

Time, t hours	0	1	2	3	4
Distance, d km					

b Find the gradient of the line.

c What does the gradient represent?

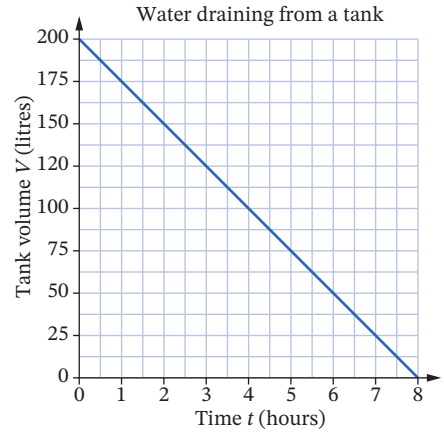
d Use the graph to find:

i the distance travelled by the car after $5\frac{1}{2}$ hours

ii the time taken for the car to travel 40 km.

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4 Water is draining out of a tank. The volume of water in the tank is shown by this graph. R C



a What is the gradient and vertical intercept of the line?

b What does the vertical intercept represent?

c What does the gradient represent?

d Why is the gradient negative?

e Find the equation of the line using the variables V and t.

5 Vinay is an emergency electrician charges a $100 callout fee plus $75 per hour thereafter. This can be represented by the equation $C = 75t + 100$ where C represents the cost in dollars and t represents the number of hours. R C

a Draw a graph to represent this equation.

b Find the gradient and vertical intercept of the line from the graph and the equation.

c Use the graph or equation to find:

i the charge for a 4-hour job

ii the time taken if the charge was $700.

6 Gabbi bought a new car for $28 000. Its value depreciates at a constant rate of $4000 per year. R C

a Copy and complete this table, then graph the values.

Number of years, n	0	1	2	3
Value, $V	28 000	24 000		

b Extend your graph to find after how many years the car will be worth nothing.

c Find the gradient and vertical intercept of the line.

d What do the gradient and vertical intercept of the line represent?

e Find the equation of the line using V and n.

7 Gabbi's car has a fuel consumption of 8 L/100 km. This means the car uses 8 litres of fuel for every 100 km travelled.

a Copy and complete this table, then graph the values. R C

Distance travelled, d km	100	200	300	400	500
Petrol used, P L					

b Find the gradient and vertical intercept of the line.

c What does the gradient represent?

d Find the equation of the line.

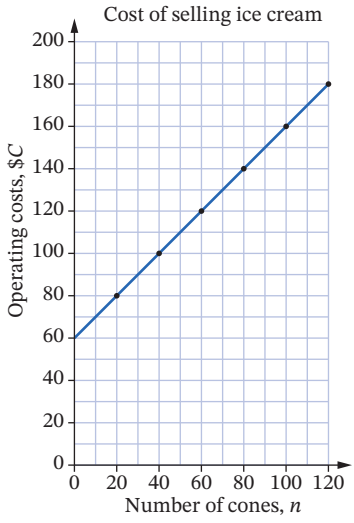
e Use the graph to find the amount of petrol used to travel 350 km.

f Verify your answer in **e** using the equation of the line.

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8 Year 10 sells ice cream at lunchtime as a fund raiser. This graph shows how much it *costs* to set up a stall and produce the ice cream cones. R C



a Before any ice cream is sold, how much does it cost to set up the stall? What costs would this represent?

b By calculating the gradient of the line, determine how much it costs to make each ice cream cone.

c Find the equation of the line.

d Calculate the total cost of making 100 ice cream cones, including the set-up costs.

e If each cone is sold for $5, how much profit would be made from selling 100 ice cream cones?

9 Pavla is saving for her end-of-school holiday. She has her birthday money of $500 and plans to add $50 to that amount each week from her part-time job for the next 2 years. R C

a Copy and complete this table of her savings, then graph these values.

Number of weeks (w)	0	10	20	30	40	50
Savings ($S)						

b What is the vertical intercept and what does it represent?

c What is the gradient of the line and what does it represent?

d Write the equation of the line.

e Use the equation to calculate how much Pavla will save after:

i 27 weeks? **ii** 60 weeks? **iii** 2 years?

d If her goal is $4500, how whole weeks will it take to achieve this amount?

10 The formula $F = 1.8C + 32$ is used to convert temperatures in degrees Celsius (°C) to degrees Fahrenheit (°F). R C

a Copy and complete this table and draw a conversion graph.

Celsius, C	–15	–10	0	10	15
Fahrenheit, F					

b Find the gradient and vertical intercept of the line using the graph and formula.

c Use the formula to convert:

i 6°C to °F **ii** 86°F to °C.

Verify your answers using the graph.

11 Kembla borrowed $1200 from his uncle to buy a kayak. He agrees to pay back $25 a week. R C

a Write an equation to show how much money (A) Kembla still *owes* after n weeks.

b Use the equation to find the amount owing after:

i 35 weeks **ii** 45 weeks.

c Graph the equation.

d Why is the gradient of the line negative?

e Use the graph to verify the vertical intercept and the gradient.

f After how many weeks will the loan be repaid?

2.08 Equations of parallel and perpendicular lines

Video
Equations of parallel lines

PATH

Worksheets
Writing equations of lines

Puzzles
Linear equations match-up

Equations of parallel lines

Example 21

Find the equation of the line parallel to $y = 8 - 3x$ that passes through the point $(-1, 6)$.

Solution

For $y = 8 - 3x$ (or $y = -3x + 8$), the gradient is $m = -3$.

A line parallel to $y = 8 - 3x$, will also have $m = -3$.

Using the point–gradient formula $y - y_1 = m(x - x_1)$ with $m = -3$ and point $(-1, 6)$:

$y - 6 = -3[x - (-1)]$

$= -3(x + 1)$

$= -3x - 3$

$y = -3x + 3$

OR: Using the gradient-intercept equation $y = mx + c = -3x + c$, where c is a constant.

To find the value of c, substitute the point $(-1, 6)$ into the equation:

$y = -3x + c$

$6 = -3 \times (-1) + c$ $\quad x = -1, y = 6$

$6 = 3 + c$

$c = 3$

$\therefore$ The equation is $y = -3x + 3$.

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Example 22

Find the equation of the line perpendicular to $3x - 4y + 6 = 0$ that passes through the point $(5, 4)$.

Solution

To find the gradient of $3x - 4y + 6 = 0$, first convert it to the form $y = mx + c$:

$3x - 4y + 6 = 0$

$3x + 6 = 4y$

$4y = 3x + 6$

$y = \frac{3x+6}{4}$

$y = \frac{3}{4}x + \frac{3}{2}$ $\quad y = mx + c$

$\therefore$ Gradient $= \frac{3}{4}$

$\therefore$ Gradient of perpendicular line $= \frac{-1}{\left(\frac{3}{4}\right)} = -\frac{4}{3}$ $\quad$ The negative reciprocal of $\frac{3}{4}$

Using the point–gradient formula $y - y_1 = m(x - x_1)$

$y - 4 = -\frac{4}{3}(x - 5)$

$3(y - 4) = -4(x - 5)$

$3y - 12 = -4x + 20$

$4x + 3y - 32 = 0$ $\quad$ In general form

OR: Using $y = mx + c$, the equation of this line is $y = -\frac{4x}{3} + c$

To find c, substitute $(5, 4)$ into the equation.

$4 = \left(-\frac{4}{3}\right) \times 5 + c$ $\quad x = 5, y = 4$

$c = \frac{32}{3}$

$\therefore$ The equation is $y = -\frac{4x}{3} + \frac{32}{3}$ or $y = \frac{-4x+32}{3}$ or

in general form: $4x + 3y - 32 = 0$

Video
Equations of perpendicular lines

2.08

PATH

EXERCISE 2.08 ANSWERS ON P. 613

Equations of parallel and perpendicular lines U F R C

EXAMPLE 21

1 Find the equation of the line that is parallel to: R C

a $y = 2x + 9$ and has a y-intercept of 4

b $y = 3x$ and has an x-intercept of -2

c $y = 5 - \frac{x}{2}$ and passes through $(-1, 6)$.

d $2x - y = 6$ and passes through $(5, -2)$

e $y = -5x - 8$ and passes through the midpoint of $(3, -10)$ and $(-5, -6)$

f $2y = x - 3$ and passes through $(6, -7)$.

EXAMPLE 22

2 Find the equation of a line that is perpendicular to: R C

a $y = \frac{x}{2}$ and has a y-intercept of -2

b $y = -5x$ and has an x-intercept of 1

c $y = 3x - 1$ and passes through the x-axis at 4

d $y = \frac{x-6}{3}$ and passes through $(1, -6)$

e $x + y - 6 = 0$ and passes through $(-4, 2)$

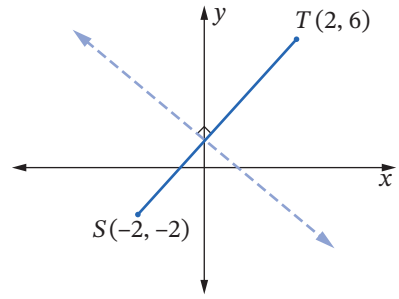
f $3x - y - 9 = 0$ and passes through $(-10, -7)$

3 **a** Find the gradient of interval ST in the diagram.

b Find the midpoint of ST.

c The dotted line is perpendicular to ST and passes through its midpoint. What is its gradient?

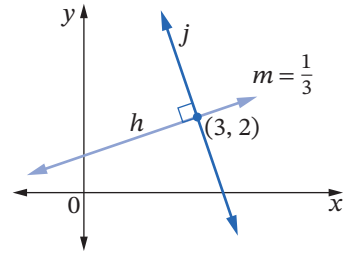
d Find the equation of the dotted line, in the form $y = mx + c$.



4 **a** Find the equation of line h in the diagram.

b Find the gradient of line j (which is perpendicular to line h).

c Find the equation of line j. R C



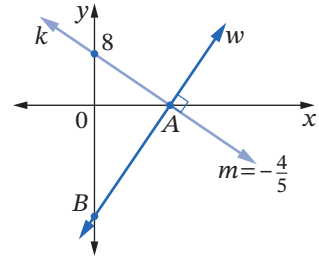
5 **a** Find the equation of line k.

b Find the coordinates of point A.

c Find the gradient of line w.

d Find the equation of line w.

e Find the coordinates of point B. R C



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Coordinate geometry proofs

2.09

A variety of problems can be solved by applying coordinate geometry methods, including proving geometrical properties of triangles and quadrilaterals.

PATH

2.09

Worksheet
Geometry problems using coordinates

Example 23

Lines k and l are shown in the diagram. Find:

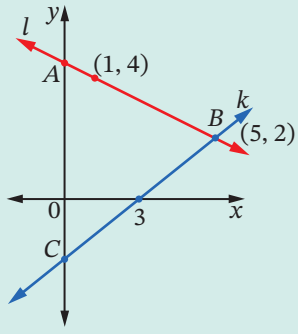
a the equation of line k

b the equation of line l

c the coordinates of point A

d the coordinates of point C

e the area of the triangle ABC.



Solution

a Line k passes through (3, 0) and (5, 2).

$$m = \frac{2-0}{5-3}$$
$$= \frac{2}{2}$$
$$= 1$$

Using the point–gradient formula $y - y_1 = m(x - x_1)$:

$$y - 2 = 1(x - 5)$$
$$= x - 5$$
$$y = x - 3$$

(or $x - y - 3 = 0$ in general form)

b Line l passes through (1, 4) and (5, 2).

$$\therefore m = \frac{2-4}{5-1}$$
$$= \frac{-2}{4}$$
$$= -\frac{1}{2}$$

$$y - 4 = -\frac{1}{2}(x - 1)$$
$$2(y - 4) = -1(x - 1)$$
$$2y - 8 = -x + 1$$
$$x + 2y - 9 = 0$$

c A is the y-intercept of line l.

Substitute $x = 0$ into $x + 2y - 9 = 0$.

$$0 + 2y - 9 = 0$$
$$2y = 9$$
$$y = \frac{9}{2}$$
$$= 4.5$$

$\therefore A$ is (0, 4.5)

d C is the y-intercept of line k.

The y-intercept of $y = x - 3$ is -3.

$\therefore C$ is (0, -3)

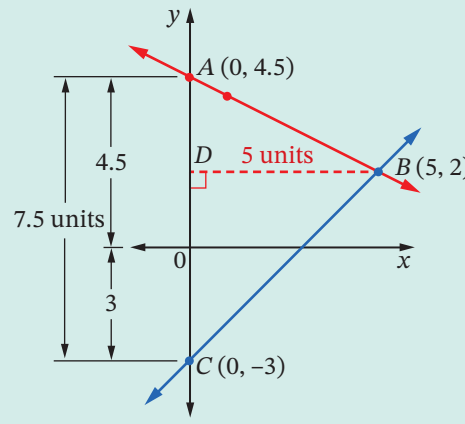
PATH

e Area of $\triangle ABC = \frac{1}{2} \times$ base $\times$ height

$= \frac{1}{2} \times AC \times BD$ $\quad AC = 4.5 + 3 = 7.5$

$= \frac{1}{2} \times 7.5 \times 5$

$= 18.75$ units^2



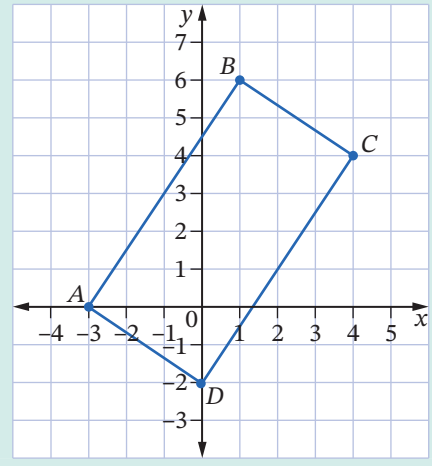
Example 24

$A(-3, 0)$, $B(1, 6)$, $C(4, 4)$ and $D(0, -2)$ are the vertices of a rectangle.

a By finding the lengths of AC and BD, show that the diagonals of the rectangle are equal.

b Find the midpoints of the diagonals AC and BD.

c Show that the diagonals of the rectangle bisect each other.



Solution

a $A(-3, 0)$, $C(4, 4)$

$AC = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$= \sqrt{[4 - (-3)]^2 + (4 - 0)^2}$

$= \sqrt{65}$

$B(1, 6)$, $D(0, -2)$

$BD = \sqrt{(0 - 1)^2 + (-2 - 6)^2}$

$= \sqrt{65}$

$\therefore AC = BD$

$\therefore$ The diagonals are equal.

b Midpoint of $AC \equiv \left(\frac{-3 + 4}{2}, \frac{0 + 4}{2}\right) \equiv \left(\frac{1}{2}, 2\right)$

Midpoint of $BD \equiv \left(\frac{1 + 0}{2}, \frac{6 + (-2)}{2}\right) \equiv \left(\frac{1}{2}, 2\right)$

c The midpoints of both diagonals are the same, so the diagonals bisect each other.

Coordinate geometry proofs U F PS R C

PATH

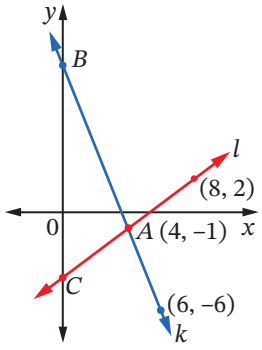
EXAMPLE 23

2.09

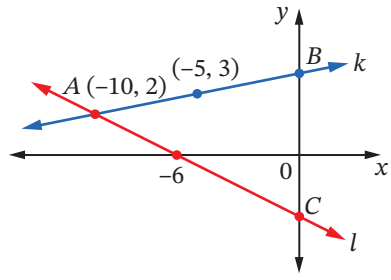
1 For each graph, find: R

- **i** the equation of line k
- **ii** the equation of line l
- **iii** the coordinates of point B
- **iv** the coordinates of point C
- **v** the area of $\triangle ABC$.

a

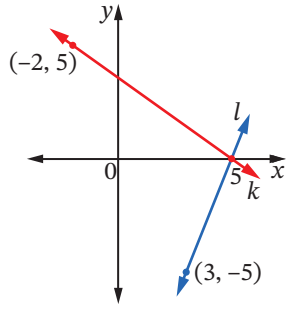


b



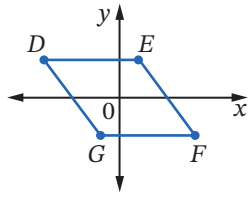
2 For this graph, find: R

- **a** the equation of line l
- **b** the equation of line k
- **c** w if the point $(7, w)$ lies on l
- **d** t if the point $(t, 11)$ lies on k



EXAMPLE 24

3 The vertices of a rhombus are $D(-4, 2)$, $E(1, 2)$, $F(4, -2)$ and $G(-1, -2)$. R C



- **a** Show that all sides of the rhombus are equal.

> **'Show that'** means all working out must be provided to fully explain your answer.

- **b** By finding their gradients, show that the opposite sides of the rhombus are parallel.
- **c** Show that the diagonals DF and GE of the rhombus cross at right angles.
- **d** Find the midpoints of the diagonals DF and GE. Do the diagonals bisect each other? Give reasons.
- **e** List the properties of a rhombus that have been demonstrated in this question.

PATH

4 A quadrilateral has vertices $P(-7, 2)$, $Q(2, -7)$, $R(5, -4)$ and $S(-4, 5)$. R C

- **a** Draw a diagram showing the given information.
- **b** Find the lengths of PR and QS in surd form.
- **c** Find the midpoints of PR and QS.
- **d** Is PR perpendicular to QS? Why?
- **e** What type of quadrilateral is $PQRS$? Explain.

5 A quadrilateral has vertices $C(2, 6)$, $D(-5, 2)$, $E(-1, -5)$ and $F(6, -1)$. R C

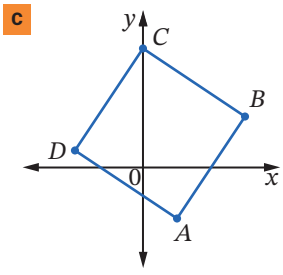
- **a** Draw a diagram showing the given information.
- **b** Find the length of each diagonal.
- **c** Find the midpoint of each diagonal.
- **d** Show that the diagonals are perpendicular.
- **e** What type of quadrilateral is $CDEF$? Explain.

6 A quadrilateral has vertices $B(1, 7)$, $C(-5, 2)$, $D(2, -2)$ and $E(8, 3)$. R

- **a** Find the length of each side.
- **b** Find the gradient of each side.
- **c** Find the midpoint of each diagonal.
- **d** What type of quadrilateral is $BCDE$?

7 A square has vertices $A(2, -3)$, $B(6, 3)$, $C(0, 7)$ and $D(-4, 1)$. R C

- **a** Show that its diagonals are equal.
- **b** Show that its diagonals bisect each other at right angles.
- **c** Hence explain why $ABCD$ is a square.



8 Show that the diagonals of the rhombus with vertices $K(4, 2)$, $L(-1, 4)$, $M(1, -1)$ and $N(6, -3)$ bisect each other at right angles. PS R C

9 Show that $W(-5, -4)$, $X(4, -1)$, $Y(6, 6)$ and $Z(-3, 3)$ are the vertices of a parallelogram by finding the gradients of each side and showing that opposite sides are parallel. R C

10 $J(-3, 0)$, $K(3, -2)$, $L(1, 3)$ and $M(-5, 5)$ are the vertices of a quadrilateral. R C

- **a** Find the gradient of each side.
- **b** What type of quadrilateral is $JKLM$? Explain.

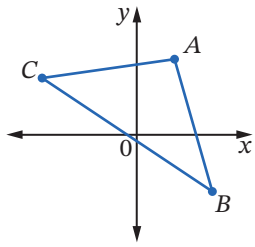
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PATH

2.09

11 $A(2, 4)$, $B(4, -3)$ and $C(-5, 3)$ are the vertices of a triangle. R C



a X and Y are the midpoints of AB and AC respectively. Find the coordinates of X and Y.

b Find the gradient of XY and CB. Is it true that $XY \parallel CB$?

c Find the lengths of XY and CB and, hence, show that $CB = 2XY$.

12 $C(-7, -6)$, $N(1, -3)$, $T(4, 5)$ and $W(-4, 2)$ are the vertices of a quadrilateral. PS R C

a Show that its diagonals bisect each other at right angles.

b What type of quadrilateral is $CNTW$? Explain.

13 What type of a quadrilateral is formed by the points $H(-6, 2)$, $I(6, -4)$, $J(4, 2)$ and $K(-2, 5)$? PS R C

14 Show that the points $S(5, -6)$, $T(6, 0)$, $W(-6, 2)$ and $X(-7, -4)$ are the vertices of a rectangle. PS R C

Power plus ANSWERS ON P. 614

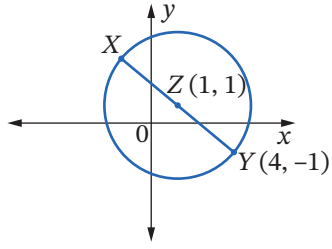
1 A line is drawn through the points $A(0, -2)$ and $B(3, 0)$. The x-coordinate of a point C on AB is 9. Find:

a the equation of AB **b** the y-coordinate of C.

2 The point $(-1, 6)$ lies on the line $kx + 3y - 13 = 0$, where k is a constant number. Find k.

3 $Z(-1, 3)$ is the midpoint of the interval joining $A(-4, 7)$ and B. Find the coordinates of B.

4 The circle has XY as a diameter and centre Z. What are the coordinates of X?



5 Show that the points $(4, 2)$, $(10, -4)$ and $(1, 5)$ are collinear.

6 **a** Find the gradient of any line parallel to $3x + 2y = 4$.

b Find the equation of the line that passes through the point $(0, -1)$ and is parallel to $3x + 2y = 4$.

7 $A(-1, 4)$, $B(4, 6)$, $C(2, 7)$ and D are the vertices of a parallelogram. Find the coordinates of D.

Foundation Standard Complex

CHAPTER 2 REVIEW

Quiz
Language of maths 2

Puzzle
Graphing lines crossword

Language of maths

axes	distance	exact answer	general form
gradient	gradient–intercept form	horizontal	interval
length	linear equation	midpoint	parallel
perpendicular	point-gradient form	reciprocal	rise
run	surd	vertical	vertical intercept
x-axis	x-intercept	y-axis	y-intercept

1 What is the difference between the y-axis and the y-intercept?

2 When finding the length of an interval on a number plane, what is meant by an **exact** answer?

3 What measurement is the fraction given by the vertical rise of a line divided by the horizontal run?

4 What is the everyday meaning of the word **intercept**? Look it up in a dictionary.

5 What is the property of the gradients of perpendicular lines?

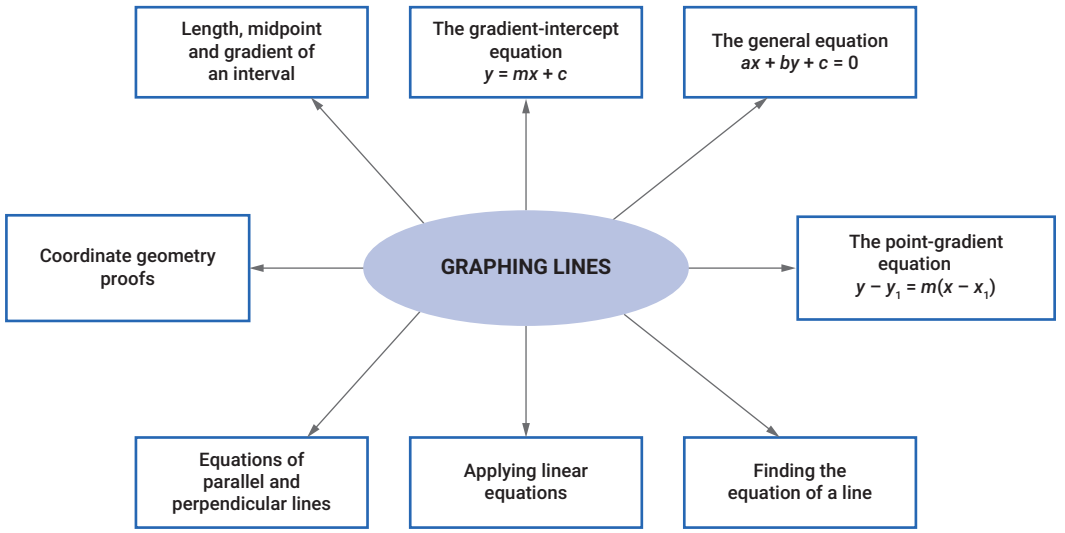
6 What form of the linear equation is $ax + by + c = 0$?

Worksheet
Mind map: Graphing lines (Advanced)

Topic summary

- How can you find the gradient of a line?
- When do you use the formula $y - y_1 = m(x - x_1)$?
- How can you test whether a pair of lines are parallel?
- What parts of this topic did you find difficult?

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



TEST YOURSELF 2

ANSWERS ON P. 614

Quiz
Test yourself 2

1 An interval is formed by joining the points $K(5, 6)$ and $L(-7, 2)$. 2.01

a Find, correct to one decimal place, the length of interval KL.

b Find the midpoint of KL.

c Find the gradient of KL.

2 The vertices of a quadrilateral $HJKL$ are $H(-8, -5)$ $J(-1, -2)$ $K(2, 5)$ $L(-5, 2)$. 2.01

a Find the exact length of the sides of the quadrilateral.

b Find the gradient of each side of $HJKL$.

c Find the exact length of the diagonals HK and JL.

d What type of quadrilateral is $HJKL$?

PATH

3 Find, correct to the nearest degree, the angle of inclination of a line with gradient: 2.01

a 3 **b** $\frac{5}{4}$ **c** -1 **d** $-\frac{2}{3}$

4 A line passes through the points $V(8, -1)$ and $W(10, -2)$. What is the gradient of a line: 2.02

a parallel to VW? **b** perpendicular to VW?

5 Graph each linear equation on a number plane. 2.03

a $y = -5x - 1$ **b** $x + 2y = 16$ **c** $3x + 4y - 12 = 0$

6 Which point lies on the line of $3x + y = 2$? Select the correct answer **A**, **B**, **C** or **D**. 2.03

A $(1, 0)$ **B** $(2, 4)$ **C** $(-1, 5)$ **D** $(-1, -5)$

7 What is the equation of the line through $(-2, 3)$ and parallel to the x-axis? Select **A**, **B**, **C** or **D**. 2.03

A $x = -2$ **B** $x = 3$ **C** $y = -2$ **D** $y = 3$

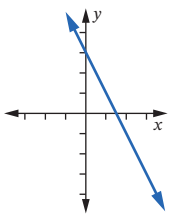
8 Write the gradient, m, and y-intercept, c, for each linear equation. 2.03

a $y = 2x - 10$ **b** $y = 4x + 3$ **c** $y = \frac{4 - 3x}{8}$

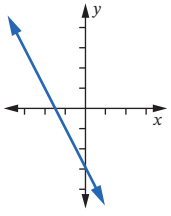
9 Match each equation to its graph below. 2.03

a $y = 2x - 3$ **b** $y = -2x - 3$ **c** $y = -2x + 3$ **d** $y = 2x + 3$

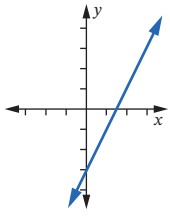
A



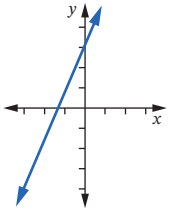
B



C



D



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10 Convert each equation to general form $ax + by + c = 0$.

2.04

a $y = 3x + 5$ **b** $y = \frac{2x}{5} - 10$ **c** $x = 3y + 6$

11 Rewrite each equation in the form $y = mx + c$, then state the gradient, m, and the y-intercept, c.

2.04

a $x - y + 2 = 0$ **b** $2x - 8y + 8 = 0$ **c** $3x + y - 9 = 0$

12 Find the x- and y-intercepts of each line in Question **11**.

2.03

13 Find in general form the equation of a line that passes through the point:

2.05

a (5, 5) and has a gradient of –3 **b** (–1, 8) and has a gradient of $\frac{2}{3}$

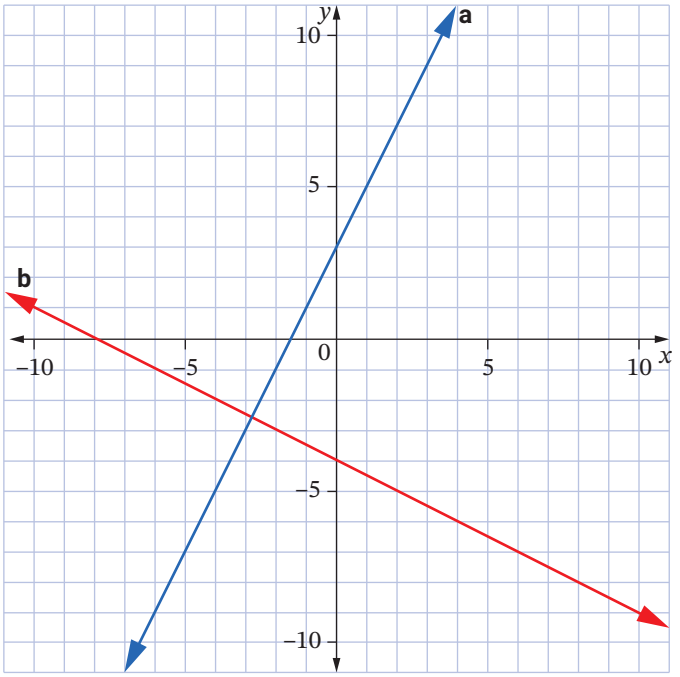
14 Find in general form the equation of a line that passes through the points:

2.05

a (10, 2) and (5, –1) **b** (–6, 3) and (–2, –1).

15 Find the equation of each line and show that they are perpendicular.

2.06



Foundation Standard Complex

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16 Seth's Self Storage charges an upfront fee of $50 plus $75 a month thereafter to store items. 2.07

a Copy and complete this table of values, then graph the values.

Number of months, n	0	6	12	18	24
Cost, $C					

b Find the gradient of the line.

c Find the vertical intercept of the line.

d Find the equation of the line.

e Use the equation or the graph to find:

i the cost of storing items for 5 months

ii the number of months that items can be stored if the cost is $650.

17 Find the equation of a line that is: 2.08

a parallel to $y = 3x + 1$ and passes through the x-axis at 2

b perpendicular to $y = \frac{x}{2}$ and passes through the origin.

PATH 2.08 2.09

18 The line $3x - 8y + 10 = 0$ and another line intersect at right angles at the point (10, 5). Find the equation of the other line.

19 $L(-1, 2)$, $M(4, 5)$, $N(1, 10)$ and $P(-4, 7)$ are the vertices of a quadrilateral. Show that $LMNP$ is a square.

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